10.3 Classification of Differential Equations

Definition 10.4: (Ordinary Differential Equation)

If a differential equation contains only ordinary derivatives of one or more functions with respect to a single independent variable, it is said to be an **Ordinary Differential Equation (ODE)**.

Definition 10.5: (Partial Differential Equation)

An equation involving only partial derivatives of one or more functions of two or more independent variables is called a **Partial Differential Equation (PDE)**.

For instance, let y denote the unknown function and x be independent variable. Then $\frac{dy}{dx}+2y=e^{-x}$, $\frac{d^2y}{dx^2}-\frac{dy}{dx}-5y=0$ and $\frac{dx}{dt}+\frac{dy}{dt}=3x-4y$ are some examples of ordinary differential equations.

For instance, $\frac{\partial u}{\partial y}=-\frac{\partial u}{\partial x}$, $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0$ and $\frac{\partial^2 u}{\partial x^2}=\frac{\partial^2 u}{\partial t^2}-2\frac{\partial u}{\partial t}$ are some examples of partial differential equations.

In this chapter, we discuss ordinary differential equations only.

Ordinary differential equations are classified into two different categories namely **linear** ordinary differential equations and **nonlinear** ordinary differential equations.

Definition 10.6

A general linear ordinary differential equation of order *n* is any differential equation that can be written in the following form.

$$a_n(x)y^{(n)}+a_{n-1}(x)y^{(n-1)}+\cdots+a_1(x)y'+a_0y=g(x) \qquad \ldots (1)$$

where the coefficients $a_n(x)\neq 0, a_0(x), a_1(x), \ldots, a_{n-1}(x)$ and $g(x)$ are any function of independent variable x (including the zero function)

Note

(1) The important thing to note about linear differential equations is that there are no products of the function, $y(x)$, and its derivatives and neither the function nor its derivatives occur to any power other than the first power.

(2) No transcendental functions – (trigonometric or logarithmic etc) of y or any of its derivatives occur in differential equation.

(3) Also note that neither the function nor its derivatives are "inside" another function, for instance, $\sqrt{y'}$ or $e^{y'}$.

(4) The coefficients $a_0(x), a_1(x), \ldots, a_{n-1}(x)$ and $g(x)$ can be zero or non-zero functions, or constant or non-constant functions, linear or non-linear functions. Only the function, $y(x)$, and its derivatives are used in determining whether a differential equation is linear.

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